Supplementary Methods and Results

Waveform Quality, Confidence, and AF-Case Retention in Single-Lead ECG Selective

Classification

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Reproduction and interpretation

This supplement describes both the fixed main experiment and separately specified follow-up explorations. Tables have been created from existing scientific records. Full condition tables, regression coefficients, predictions, source catalogs, masks, and bootstrap samples have been saved within the technical repository. The original primary finding does not become nullified through this exploratory ablation procedure.

The computational setup has run on Python 3.12.14 running on Linux x86_64, using NumPy 2.3.5, pandas 2.2.3, SciPy 1.17.0, scikit-learn 1.8.0, and WFDB 4.3.0. Dependencies for the CPU version of Torch have been listed separately in the repository. Platform-independent bit-level reproducibility cannot be guaranteed.

Signal processing and score orientation

In our current study, let x represent a centered ECG signal and n represent a centered resampled noise. The resulting corrupted signal becomes x+[RMS(x)/RMS(n)]×10^-(SNR/20)×n, where RMS indicates the root-mean-square calculated across all 1,000 samples. Any signals resulting from source windows of zero power fail validation instead of being ignored in the data recording process. This addition-to-noise scheme does not conform to the native total SNR or to the WFDB QRS amplitude convention.

For the classification task, the input data points are normalized to be centered, followed by dividing by a fixed training-only standard deviation of 0.1618611220663296 mV based solely upon the training data. Quality metrics are determined before normalization in terms of physical units of millivolts. Increasing raw-confidence error score or learned error score implies an expected higher error rate and should therefore take precedence as the problem of interest. Specifically, the raw-confidence error score is set to 1-max(p,1-p) rather than to the absolute distance between the chosen decision boundary of 0.562615.

Dependence and uncertainty

Participants may supply multiple recordings associated with different rhythmic categories or multiple native/artifact tiers. Tally counts per class or artifact category do not need to add up to the overall tally count. Positive-participant gates refer to unique participants with at least one positive recording entry. Corruption instances refer to repeated sampling, not unique participants.

Primary bootstrap samples include all participant recordings as well as severity classes. Multiplicities accurately simulate resampled and re-ranked boundary values. The primary confidence interval includes no undefined replicate entries. Certain exploratory stratifications feature undefined balanced error replicates owing to deferment of elimination of a class; such numbers are reported in Supplementary Table S6. Confidence intervals use only valid replicates if the percentage of undefined replicates is no greater than 1%. Such confidence intervals still condition on model fits and the reused noise library.

Table S1. Development-only checkpoint comparators

Model	AUROC	AP	Sensitivity %	Specificity %
CNN	0.9898	0.8668	90.00	98.56
RR logistic	0.7251	0.1351	90.00	52.65
Constant negative	0.5000	0.0638	0.00	100.00

Fold 7 consists of 1,410 ECGs, out of which 90 exhibit AF. These data are intended for checkpoint/cutoff choice and are not a standalone validation/test dataset. Features related to RR intervals are the total number of beats, median of RR, RR standard deviation, RR coefficient of variation, root mean square successive differences (RMSSD), and percentage of successive RR differences >50 ms (pNN50), determined automatically from XQRS detection.

Table S2. Quality measurement definitions

Measurement	Definition
AC standard deviation	Population standard deviation of centered samples, in mV.
Peak-to-peak range	Maximum minus minimum sample, in mV.
Near-flat fraction	Fraction of adjacent absolute differences <= 0.000001 mV.
Relative adjacent RMS	Root mean square adjacent difference / AC standard deviation.
Low-band ratio	Welch power 0.05-0.5 Hz / power 0.05-40 Hz.
High-band ratio	Welch power 20-40 Hz / power 0.5-40 Hz.
Pearson kurtosis	Pearson kurtosis with finite-sample bias correction (fisher=False, bias=False), not excess kurtosis.

Welch Method Parameters

The Welch method used Hann window function with nperseg = 256, noverlap = 128, with constant detrending and spectral density scaling. The integration process employed the trapezoid rule, taking into account the boundary powers estimated via linear interpolation. Where the denominator was close to zero (≤1×10^-12), the output contained NaNs and the appropriate missingness flag. Median imputation and scaling could only take place on fold-8 data. The initial 15-input quality ranker had 7 quality metrics, 7 flags, and 1 prediction class, while the alternative quality descriptor was made up of 14 inputs.

Table S3. Denominators for selective prediction

Measure	Numerator / denominator
Overall coverage	All retained records / all original records
AF-case retention	Retained AF-positive records / all original AF-positive records
Retained sensitivity	Retained true-positive classifications / retained AF-positive records
Retained specificity	Retained true-negative classifications / retained negative records
True-positive preservation	Retained true-positive classifications / all original AF-positive records
Ordinary retained error	Retained incorrect classifications / all retained records
Balanced retained error	0.5 x (retained FN / retained positives + retained FP / retained negatives)

AF-positive ECGs that had been designated as negative are termed “retained FN,” while negative ECGs that had been labeled as positive are termed “retained FP.” No subsequent action was assumed to occur on these deferrals.

Table S4. All seven exploratory ablations

Ablation	Reference	Diff. pp	95% interval pp	AF retention %
confidence no class	confidence adjusted	+0.90	-2.50 to +2.35	9.27-41.72
quality no class	quality only	-33.90	-37.26 to -30.42	85.43-89.40
combined no class	combined	-0.42	-2.54 to +2.39	9.27-43.71
combined no amplitude	combined	+1.37	-1.09 to +2.71	9.93-48.34
combined no spectral	combined	-0.50	-0.83 to +1.34	9.27-49.67
combined no temporal	combined	+0.00	-0.54 to +0.19	9.27-48.34
combined no kurtosis	combined	-0.09	-0.89 to +1.67	9.27-49.01

Classification parameters: 24/12/6/0 dB; nominal coverage of 90%. Corresponding missingness flags were removed only when their associated quality-feature groups were removed. All seven variants are provided. Neither multiplicity correction nor designation as a main model has been performed.

Table S5. Additional development seeds

Seed	Epoch	AF cutoff	Diff. pp	95% interval pp	AF retention %
20261016	7	0.438684	-1.18	-3.58 to +3.55	10.60-34.44
20261017	8	0.213014	-1.93	-4.55 to +0.59	8.61-66.89

Each seed performs iterations of training, selecting checkpoints or cutoffs, training the ranking model, and employing unlabeled policies within their appropriate context. In both cases, the initial test set remains the same; hence, no independent replications in patients occur. Both fail to satisfy the 80% AF-case retention criterion at every noise severity.

Table S6. Cohort and subgroup sensitivity

Group	ECGs	AF+ people	Diff. pp	95% interval pp	Undefined draws
exclude mixed rhythms	2071	127	-0.61	-2.63 to +0.95	0
one earliest per participant	1812	115	+0.23	-0.88 to +0.99	2
age18 64	1097	25	Not estimated	Insufficient positives	-
age65plus	988	104	+0.55	-0.44 to +2.01	0
recorded sex0	1080	66	-1.71	-3.54 to +2.08	6
recorded sex1	1005	63	-1.16	-3.75 to +1.90	2
native artifact annotated	505	40	+0.42	-4.23 to +2.84	1
no native artifact annotation	1580	96	+0.11	-1.97 to +1.24	7

For each numeric analysis, the members of each subgroup are reordered based on 90% coverage. These numbers do not describe one globally retained set. All confidence intervals intersect zero, and each of the combined AF-retention guards fail. The age 65+ group includes provider deidentified oldest-age codes. Sex remained coded 0/1 according to provider codebooks. Participant pools overlap among different artifact categories. At least 30 unique positive and 30 unique negative participants were required.

Table S7. Retained confusion counts at the primary budget

SNR dB	Method	TP	FN	TN	FP	AF retention %
24	raw confidence	49	8	1813	7	37.75
24	confidence adjusted	0	14	1863	0	9.27
24	quality only	0	30	1847	0	19.87
24	combined	0	14	1863	0	9.27
12	raw confidence	50	10	1809	8	39.74
12	confidence adjusted	0	16	1861	0	10.60
12	quality only	0	34	1843	0	22.52
12	combined	0	17	1860	0	11.26
6	raw confidence	45	20	1803	9	43.05
6	confidence adjusted	2	27	1848	0	19.21
6	quality only	0	49	1828	0	32.45
6	combined	4	28	1843	2	21.19
0	raw confidence	51	34	1770	22	56.29
0	confidence adjusted	39	35	1790	13	49.01
0	quality only	0	71	1806	0	47.02
0	combined	36	37	1791	13	48.34

Each row has 1,877 entries. Prior to the exclusion of positives, there are 151 AF-positives and 1,934 negatives. The initial quality-based strategy utilized the predicted classification label; a row yielding zero true positives does not receive any beneficial interpretation.

Table S8. Frozen unlabeled-policy cutoffs

Method	Cutoff	Weighted coverage %	Equal-condition coverage %
Raw confidence	0.06192078	90.0088	89.4362
Confidence-adjusted	0.04276845	90.0008	89.9357
Quality + class	0.03935559	90.0008	89.2821
Combined	0.04476287	90.0008	89.7763

Fold 9 includes 2,091 ECGs and 18,819 variant sets. The unmodified, baseline-wander (bw), and muscle-artifact (ma) types each contribute equally to weighting of 1/3; each noise severity level contributes 1/12. Labels for outcomes were not used. Coverage scores refer to evaluation of the development score distribution and not test accuracy.

Source integrity and external selection

The checksum inventory for all 30,584 primary cohort header/signal files, 3,864 external header/signal files chosen from the provider pool, and six noise header/signal files was successfully validated. Specifically, the external pool includes 2,000 header selections and 1,864 qualified signal files. No signals associated with exclusions needed to be included. A static hash of 20261019| provider-relative-header-path was performed to choose the 2,000 external headers before aging/rhythming analysis; hashing is deterministic but not random sampling. For atrial fibrillation (AFIB), use SNOMED CT code 164889003. For atrial flutter, use 164890007 (which should never be labeled as AFIB).

Review budgets and saved condition tables

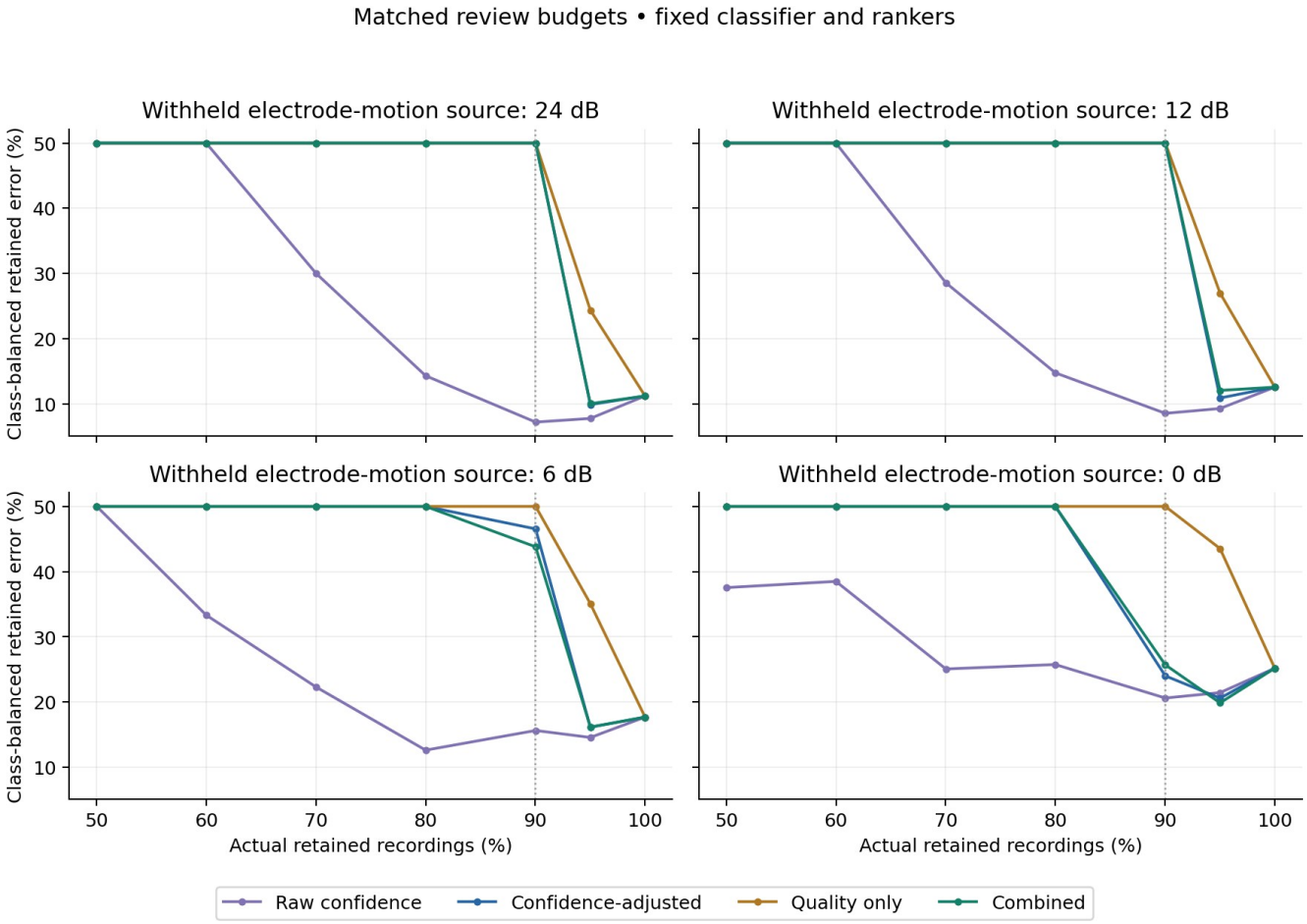
For the main experiment, we use seven matched coverages (50%, 60%, 70%, 80%, 90%, 95%, 100%) for the base case and twelve corruption scenarios. Policy settings validated for the purpose are provided in a different table. Robustness testing involves three matched coverages (80%, 90%, and 95%) along with diagnostic tests of positive cases. Full CSV tables are provided alongside the manuscript submission package to prevent the supplement from hiding any condition-level findings.

Protocol correction and freeze

It should be noted that the initial local protocol lock occurred before test acquisition. An amendment to the documentation clarified the difference between float32 and float64 accuracy levels exclusively in the SNR verification reference document. In this case, the maximum measured final SNR error was found to be 7.11 × 10^-15 dB. Neither the generated outputs nor the fitting method or cutoff values have been modified. The subsequent severity transfer and robustness criteria still apply separately. Evidence-freeze verification has to occur using the full package containing sources, not the trimmed-down analysis version, as the latter requires reconstructing the sources.

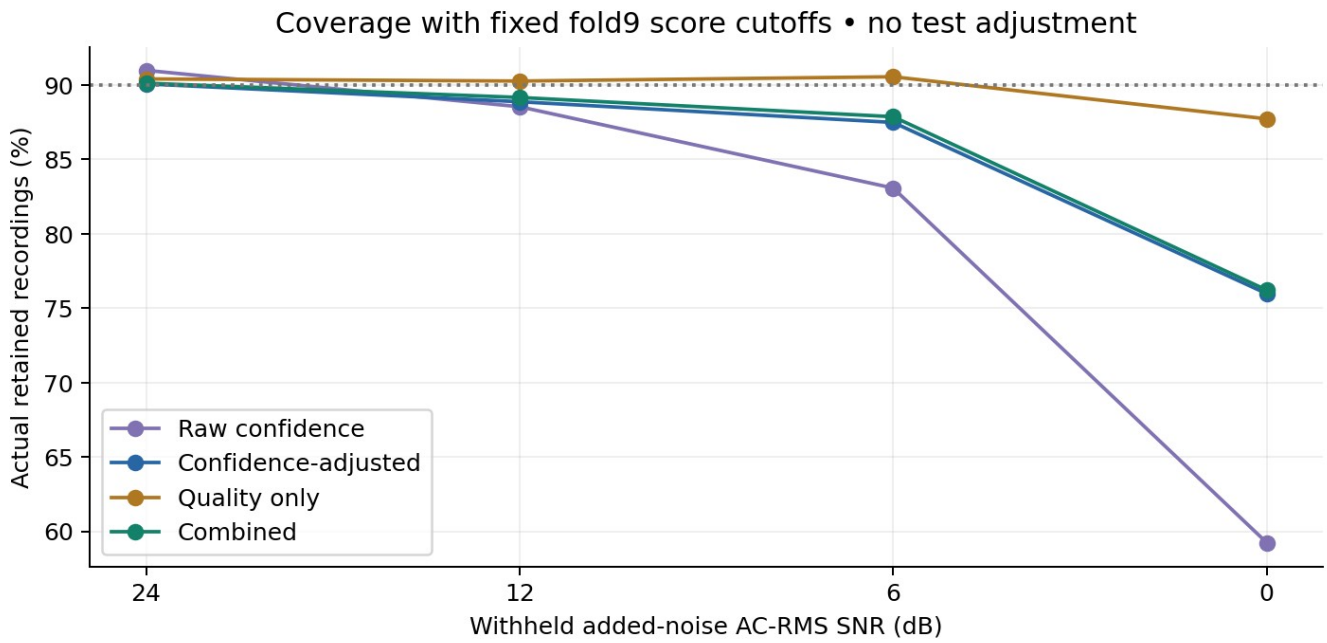
No text change in the manuscript, no improved sensitivity model, nor addition of any further supporting table affects the original key finding. For future revisions based on science, amendments along with timestamps and snapshots need to be retained. All entries in the research stage use past-tense statuses, where the latest evidence assessment defines completion. Automated internal verification cannot replace peer review, ethical approval, acceptance for publication, or authorship independent of assistance.

Figure S1. Primary balanced risk-coverage curves



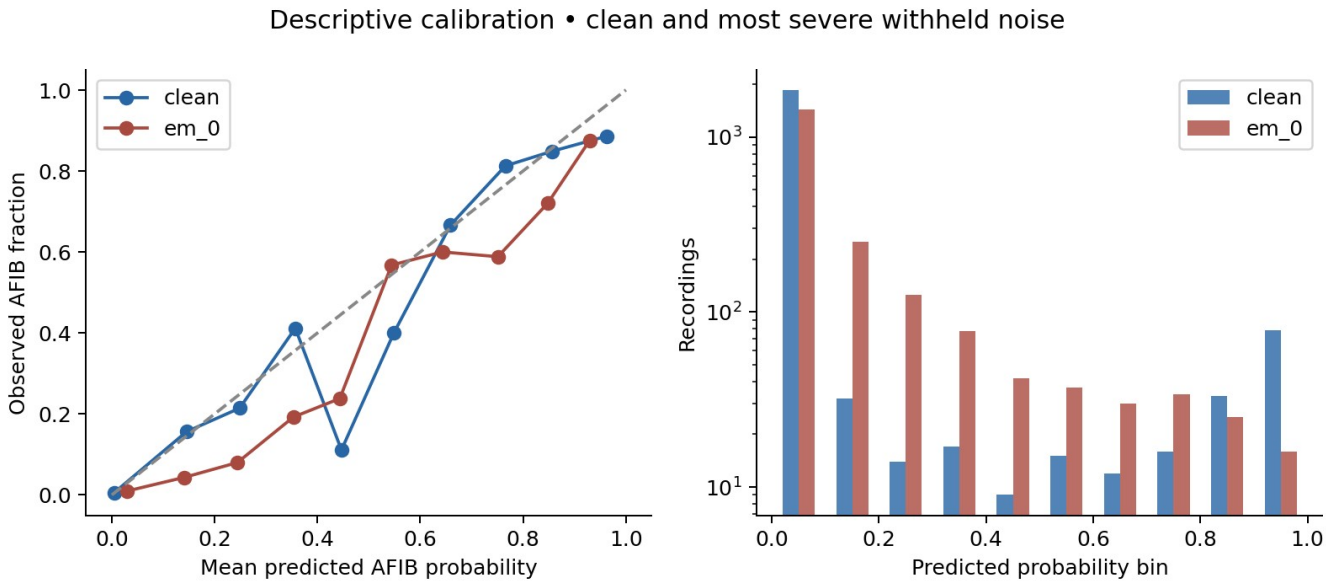
Four error-ranking methods across retained-record budgets and four em severities. Curves are conditional on the fixed classifier and noise assignments. Case-retention values must accompany any accuracy interpretation.

Figure S2. Actual coverage of frozen score policies



Policies were selected using unlabeled fold 9 scores and applied without test-outcome tuning. A validation-fixed policy is distinct from reranking each condition to a fixed retained count.

Figure S3. Classifier calibration displays



Ten fixed equal-width probability bins, with the last including 1. Bin means are descriptive; no recalibration was performed. Sparse bins and changed class prevalence constrain interpretation.